Non-Markovian Memory and Temporal Accumulation: A Conditional Thermodynamic Analysis

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Abstract

We investigate a conditional thermodynamic role for memory in open quantum systems interacting with stochastic environments. Working within stochastic and information thermodynamics, we define a *survival functional* $\mathcal{S} := \Delta F - W$ measuring the difference between non-equilibrium free energy gained and work invested by an agent.

For a fixed-Hamiltonian autonomous Davies/GKSL baseline with no measurement or feedback, we prove a restricted Markovian ceiling: $\mathcal{S} \leq 0$. This is not a universal ceiling for driven or feedback-controlled Markovian agents. We then derive an accounting identity for finite-dimensional autonomous evolution, with the interaction-energy term retained or a declared weak-coupling approximation made, expressing the survival functional through changes in system–environment mutual information and bath displacement. In models where prior interaction has supplied usable correlations, their consumption during non-Markovian backflow intervals can yield $\mathcal{S} > 0$, bounded by the initial correlation budget.

The result is therefore a conditional memory advantage, not a proof that memory or non-Markovianity is necessary or sufficient for all persistent systems. Fisher accretion and the comparison with the accessibility ordering of the Holographic Alaya-Field Framework (HAFF) are presented as heuristics. A worked spin-boson example with Lorentz–Drude spectral density illustrates how the accounting can operate in one supplied model.

Finally, using entropy rate and predictive information from computational mechanics, we study the implementation cost of a specified irreversible archive-reset model. Universal Landauer costs for merely storing or maintaining memory, and the resulting universal "Memory Catastrophe," are withdrawn; only the declared archive implementation is charged.

Keywords: non-Markovian dynamics, open quantum systems, Nakajima–Zwanzig equation, memory kernel, thermodynamic arrow of time, information backflow, entropy production, stochastic thermodynamics

1 Introduction

1.1 Context: The Problem of Persistence

A quantum system coupled to a thermal environment generically relaxes toward equilibrium. This is the content of the *zeroth crisis*: absent special structure, every open subsystem is eventually erased by thermal noise [1].

Yet the physical world contains persistent far-from-equilibrium structures—from molecular machines to living organisms—that maintain themselves against the entropic tide for timescales vastly exceeding their intrinsic relaxation times. What structural feature of their dynamics makes this possible?

The standard answer invokes free-energy input: a persistent system continuously imports low-entropy energy and exports high-entropy waste [2]. This paper asks the narrower accounting question of when pre-existing system–environment correlations can fund a later interval. It does not claim that memory distinguishes every persistent system from every non-persistent one.

1.2 Position within the Series

This paper is the first of three constituting the **T-DOME** (Thermodynamic Dynamics of Observer-Memory Entanglement) framework, the third pillar of a three-paper program.

Framework	**Question**		**Result**	**Status**
HAFF [19, 20]	How does geometry emerge?	Ocean	Conditional algebra/dictionary models	Revised
Q-RAIF [23, 24]	What algebra must an observer have?	Fish	Clifford representation as model input	Revised
T-DOME I (this work)	What can inter-epoch correlations fund?	Seed	ELV-type accounting identity; restricted ceiling	**This paper**
T-DOME II	How can a supplied archive be budgeted?	Ego	Conditional feature selection	Companion
T-DOME III	When can residual statistics track drift?	Loop	Conditional Fisher calibration	Companion

The three T-DOME papers form a conditional modelling sequence, not an irreversible theorem chain:

1. **Paper I (The Seed):** The accounting identity asks when correlations built in earlier epochs can fund a later interval. A deliberately uncompressed archive is then introduced only as motivation for a finite-budget problem.

2. **Paper II (The Ego):** Given that archive model and a supplied component representation, Paper II studies conditional feature selection under a processing budget.

3. **Paper III (The Loop):** Given a selected frame and an explicit residual likelihood, Paper III studies conditions under which a calibration controller can track drift.

The present paper addresses only the first link in this chain.

1.3 Relation to HAFF Paper F

Revised HAFF Paper F [22] treats record growth and algebra inclusion as protocol-relative orderings and does not derive a universal temporal partial order. Section 5 therefore uses it only as motivation: a memory-influence relation and a pointer-record ordering may be compared after both are supplied, but neither is proved to embed in the other.

1.4 Scope and Disclaimers

1. This work does *not* claim that memory or non-Markovianity is necessary for persistence. Continuously driven Markovian dissipative structures can persist. The narrower claim is that pre-existing inter-epoch correlations can fund a positive interval of the survival accounting functional.

2. The sign of that functional is governed by changes in initial correlation and bath-displacement resources, not by Markovianity as such. No strict comparison of Markovian and non-Markovian suprema is proved.

3. We do *not* derive the specific form of the memory kernel from first principles. The kernel is treated as a structural feature of the system-environment coupling.

4. The term "agent" is used in the control-theoretic sense (a subsystem that acts on its environment to maintain a target state) and carries no implication of consciousness, intention, or subjective experience.

5. A broader structural analogy with classical philosophical concepts of temporal persistence exists but is outside the scope of this paper.

Result-status ledger. The information–thermodynamic identity is a general finite-dimensional unitary accounting identity (GvN/THM/OK-cond, with the interaction-energy term retained or a declared weak-coupling approximation). The original time-dependent-control Markovian Ceiling is retracted; a narrower autonomous Davies-semigroup theorem is proved below (FD/I/THM/OK-cond). Fisher accretion and the HAFF bridge are heuristics. Universal Landauer costs for storing or maintaining memory and the resulting "Memory Catastrophe" are retracted; only a specified irreversible archive-reset model remains.

2 Mathematical Preliminaries

2.1 Open Quantum Systems: The Markovian Baseline

Consider a bipartite Hilbert space $\mathcal{H} = \mathcal{H}_R \otimes \mathcal{H}_E$, where R denotes the "agent" (reduced system) and E the environment. The total Hamiltonian is

$$H = H_R \otimes \mathbb{1}_E + \mathbb{1}_R \otimes H_E + \lambda\, H_{\text{int}}, \tag{1}$$

where λ parametrizes the coupling strength.

Under the Born–Markov and secular approximations, the reduced dynamics of $\rho_R(t) = \text{Tr}_E[\rho(t)]$ is governed by the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation [3, 4]:

$$\dot{\rho}_R(t) = -i[H_{\text{eff}},\, \rho_R(t)] + \sum_k \gamma_k \left(L_k \rho_R(t) L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k,\, \rho_R(t)\} \right), \tag{2}$$

with $\gamma_k \geq 0$ and Lindblad operators $\{L_k\}$.

Remark 1 (Markovian = Memoryless)**.** *The GKSL equation is* time-local*: $\dot{\rho}_R(t)$ depends only on $\rho_R(t)$, never on $\rho_R(s)$ for $s < t$. Physically, this corresponds to an environment with vanishing correlation time ($\tau_E \to 0$): the bath "forgets" its interaction with the system instantaneously. The semigroup property $\Lambda(t+s) = \Lambda(t)\Lambda(s)$ ensures complete positivity at all times but precludes any information backflow from environment to system [9].*

2.2 Beyond Markov: The Nakajima–Zwanzig Equation

When the environmental correlation time τ_E is non-negligible, the Born–Markov approximation fails. The exact reduced dynamics is captured by the Nakajima–Zwanzig (NZ) integro-differential equation [5, 6]:

$$\dot{\rho}_R(t) = -i[H_{\text{eff}},\, \rho_R(t)] + \int_0^t ds\; \mathcal{K}(t,s)\, \rho_R(s), \tag{3}$$

where $\mathcal{K}(t,s)$ is the **memory kernel**—a superoperator encoding the influence of the system's entire history on its present dynamics.

Definition 2 (Memory Kernel)**.** *The memory kernel $\mathcal{K} : [0,\infty)^2 \to \mathcal{L}(\mathcal{B}(\mathcal{H}_R))$ is the superoperator satisfying* (3)*. It encodes two types of information:*

1. ***Environmental structure:*** *the spectral density, correlation functions, and non-equilibrium features of the bath;*
2. ***Temporal reach:*** *the effective support $\tau_{\text{mem}} := \inf\{\tau : \|\mathcal{K}(t,s)\| < \epsilon \;\forall\; t - s > \tau\}$, the "memory depth."*

The Markovian limit corresponds to $\mathcal{K}(t,s) \to \mathcal{K}_0\, \delta(t-s)$, recovering the GKSL generator.

Remark 3 (Information Backflow)**.** *Non-Markovian dynamics admits* information backflow*: the distinguishability of two initial states, as measured by trace distance $D(\rho_1(t), \rho_2(t))$, can temporarily increase [13]. This is the operational signature of memory—the environment returns previously absorbed information to the system.*

2.3 Thermodynamic Framework

We adopt the framework of stochastic thermodynamics for open quantum systems [14]. The following conventions are fixed throughout.

Definition 4 (Thermodynamic Setup)**.**

1. ***Hamiltonian decomposition.*** *The system Hamiltonian is $H_S(t) = H_R + H_{\mathrm{ctrl}}(t)$, where H_R is the* fixed *bare Hamiltonian and $H_{\mathrm{ctrl}}(t)$ is the agent's time-dependent control protocol. The bath Hamiltonian H_E and coupling H_{int} are as in* (1)*.*

2. ***Reference state.*** *The thermal equilibrium state of the bare Hamiltonian is*

$$\rho_{\mathrm{eq}} := \frac{e^{-\beta H_R}}{Z_R}, \qquad Z_R := \mathrm{tr}\big(e^{-\beta H_R}\big), \qquad \beta := (k_B T)^{-1}. \tag{4}$$

   Since H_R is time-independent, ρ_{eq} is a well-defined, fixed reference throughout the protocol.

3. ***Non-equilibrium free energy.*** *For any state ρ of the reduced system,*

$$F(\rho) := \mathrm{tr}(\rho\, H_R) + \beta^{-1}\,\mathrm{tr}(\rho \ln \rho) = \langle H_R \rangle_\rho - \beta^{-1} S(\rho), \tag{5}$$

   where $S(\rho) = -\,\mathrm{tr}(\rho \ln \rho)$ is the von Neumann entropy. The equilibrium value is $F_{\mathrm{eq}} = -\beta^{-1} \ln Z_R$.

4. ***Free energy–relative entropy identity.***

$$D_{\mathrm{KL}}(\rho \| \rho_{\mathrm{eq}}) = \beta\big(F(\rho) - F_{\mathrm{eq}}\big) \geq 0. \tag{6}$$

   Thus D_{KL} measures the free-energy surplus in units of $k_B T$.

5. ***Work.*** *The inclusive work performed on the total system by the control protocol over $[0, \tau]$ is*

$$W[0,\tau] := \int_0^\tau \mathrm{tr}\left(\rho(t)\, \frac{\partial H_{\mathrm{ctrl}}}{\partial t}\right) dt. \tag{7}$$

6. ***Entropy-production functional.*** *The* generalised entropy production *over $[0, \tau]$ is*

$$\Sigma[0,\tau] := \beta\big(W[0,\tau] - \Delta F\big), \tag{8}$$

   where $\Delta F = F(\rho(\tau)) - F(\rho(0))$. For uncorrelated (product) initial states, $\Sigma \geq 0$ recovers the standard second-law bound. For initially correlated states, Σ can be negative*, reflecting the consumption of pre-existing correlations (see Remark 25).*

Remark 5 (Why H_R is fixed)**.** *The bare Hamiltonian H_R defines the system's energy scale and hence the reference state ρ_{eq}. The agent acts on the world through $H_{\mathrm{ctrl}}(t)$, which may be time-dependent. This separation ensures that ρ_{eq} is well-defined and time-independent for the accounting identity. It does* not *make a driven generator satisfy detailed balance with respect to that fixed state. The ceiling result below therefore uses autonomous $H_{\mathrm{ctrl}} = 0$ dynamics.*

Definition 6 (Standing Assumptions)**.** *The following minimal assumptions are in force throughout Sections 4–6 unless stated otherwise. Every main result (Lemma 19, Theorem 21, Corollary 24) relies* only *on items* (A1)–(A5) *below.*

(A1) ***Finite-dimensional bipartite system.*** $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$*, with total Hamiltonian* (1) *and global unitary evolution* $U(t) = \mathcal{T} \exp\Big(-i \int_0^t H(s)\, ds\Big)$.

(A2) ***Weak coupling.*** *The system–environment interaction satisfies* $\lambda \ll 1$ *in* (1)*, so that* $\Delta\langle H_{\rm int}\rangle = O(\lambda)$ *[1]. The first-law bookkeeping is* $W = \Delta\langle H_R\rangle + \Delta\langle H_{\rm ctrl}\rangle + \Delta\langle H_E\rangle + \Delta\langle H_{\rm int}\rangle$*. Dropping the last term is the declared weak-coupling approximation* $\Delta\langle H_{\rm int}\rangle = O(\lambda)$*; it is not an energy-conservation equation with a zero right-hand side.*

(A3) ***Fixed environmental reference.*** $\rho_E^{\rm th} := e^{-\beta H_E}/Z_E$ *is a fixed* bookkeeping *Gibbs state at inverse temperature* β*. The* actual *initial bath state* $\rho_E(0)$ *need not coincide with* $\rho_E^{\rm th}$*; when* $\rho_E(0) \neq \rho_E^{\rm th}$*, the quantity* $D_{\rm KL}(\rho_E(t)\|\rho_E^{\rm th})$ *tracks the nonequilibrium free energy stored in the bath relative to this reference. The bath Hamiltonian* H_E *is time-independent.*

(A4) ***Arbitrary initial state.*** *The total initial state* $\rho_{SE}(0)$ *is* not *required to be a product state. In particular, initial system–environment correlations* $I(S{:}E;\,0) > 0$ *and initial bath displacement* $D_{\rm KL}(\rho_E(0)\|\rho_E^{\rm th}) > 0$ *are both permitted.*

(A5) ***Regularity.*** *All quantum states appearing in the thermodynamic identities are assumed to have full rank (or are restricted to their support), so that all relative entropies* $D_{\rm KL}(\rho\|\sigma)$ *are finite.*

Remark 7 (Bookkeeping conventions)**.** *The heat absorbed by the environment is* $Q := \Delta\langle H_E\rangle = \mathrm{Tr}[\rho_E(\tau)H_E] - \mathrm{Tr}[\rho_E(0)H_E]$ *(matching Esposito* et al. *[14]). We define* $\Sigma := \beta(W - \Delta F)$ *as a* generalised entropy-balance functional*; for correlated initial conditions* Σ *need not be nonnegative (see Remark 25).*

2.4 The Survival Functional

We now define the central quantity of this paper.

Definition 8 (Survival Functional)**.** *For a reduced system* R *evolving under dynamics* Λ *over* $[0,\tau]$*, the* ***survival functional*** *is*

$$\mathcal{S}[\Lambda,\tau] := \Delta F - W[0,\tau] = \big[F(\rho(\tau)) - F(\rho(0))\big] - W[0,\tau]. \tag{9}$$

Equivalently, using (8)*,*

$$\beta\,\mathcal{S}[\Lambda,\tau] = -\Sigma[0,\tau]. \tag{10}$$

Note on nomenclature. We retain the term "survival functional" to emphasize the biological interpretation of persistence far from equilibrium; mathematically, $\mathcal{S}$ is strictly a *generalized entropy-balance functional* derived from the first and second laws.

Remark 9 (Interpretation)**.** *The survival functional has a transparent physical meaning:*

- $\mathcal{S} > 0$*: the system gained more free energy than was invested by the external protocol—a* thermodynamic profit*. The agent has extracted usable work from environmental correlations.*
- $\mathcal{S} = 0$*: the agent breaks even (reversible limit,* $\Sigma = 0$*).*
- $\mathcal{S} < 0$*: the agent paid more than it gained (the generic irreversible case).*

Under the standard second law ($\Sigma \geq 0$*),* $\mathcal{S} \leq 0$ *always. As Section 4 shows, a positive interval requires consumption of a resource present at that interval's initial time, such as system–environment correlation or bath displacement. Whether a memory kernel preserves access to that resource is a model-specific dynamical question.*

Remark 10 (Connection to Information Thermodynamics)**.** *In the Sagawa–Ueda framework [7, 8], a system under feedback control satisfies the generalized second law*

$$\Sigma \geq -I_{\text{feedback}}, \tag{11}$$

where $I_{\text{feedback}} \geq 0$ *is the mutual information gained through measurement of the system. This permits* $\Sigma < 0$ *(and hence* $\mathcal{S} > 0$*) at the expense of information. Pre-existing correlations can enter the entropy balance in a role analogous to measurement information. A memory kernel is not automatically a measurement or feedback controller; a concrete model must show that it preserves and later exposes those correlations.*

3 A Restricted Markovian Baseline

The former "Markovian Ceiling" allowed a time-dependent control Hamiltonian while using a fixed bare Gibbs state as the detailed-balance reference. A driven generator need not preserve that state, so the asserted balance law did not follow from Spohn's inequality. That theorem, its open-loop control class, and its temporal-blindness corollary are retracted. The autonomous result below is the narrower statement that survives.

3.1 Spohn's Inequality

Throughout this section we assume that the GKSL generator $\mathcal{L}$ is a *thermal Lindbladian*: it is obtained from the weak-coupling (Davies) limit of a system coupled to a single thermal bath at inverse temperature β, and satisfies **quantum detailed balance** (the KMS condition) [10, 1]. Under this assumption, the unique stationary state is the Gibbs state $\rho_{\text{ss}} = \rho_{\text{eq}}$ of (4), and the generator is self-adjoint with respect to the KMS inner product. This ensures that the entropy production rate below is well-defined and non-negative.

Definition 11 (Autonomous Davies baseline)**.** *The autonomous baseline has* $H_{\text{ctrl}} = 0$*, fixed system Hamiltonian* H_R*, and a time-independent Davies/GKSL generator* $\mathcal{L}$ *satisfying quantum detailed balance with respect to the faithful Gibbs state* $\rho_{\text{eq}} = e^{-\beta H_R}/Z_R$*. No measurement, feedback, or initial system–bath resource is included in this reduced semigroup model.*

Lemma 12 (Spohn [10])**.** *For any GKSL dynamical semigroup $\Lambda_t = e^{\mathcal{L}t}$ satisfying quantum detailed balance with unique invariant state ρ_{eq}, the entropy production rate*

$$\sigma(t) := -\operatorname{tr}\big(\mathcal{L}[\rho(t)]\,(\ln\rho(t) - \ln\rho_{\mathrm{eq}})\big) \tag{12}$$

satisfies $\sigma(t) \geq 0$, equivalently $dD_{\mathrm{KL}}(\rho(t)\|\rho_{\mathrm{eq}})/dt \leq 0$.

Proof. This follows from the contractivity of CPTP maps under quantum relative entropy [10, 1]: $D_{\mathrm{KL}}(\Lambda_t\rho\|\Lambda_t\rho_{\mathrm{eq}}) \leq D_{\mathrm{KL}}(\rho\|\rho_{\mathrm{eq}})$ for all $t \geq 0$. Differentiating at $t = 0$ yields $\sigma(t) \geq 0$. □

3.2 The Restricted Autonomous Ceiling

Theorem 13 (Restricted Autonomous Ceiling)**.** *Under Definition 11, the survival functional obeys*

$$\mathcal{S}[0,\tau] = F(\rho(\tau)) - F(\rho(0)) \leq 0. \tag{13}$$

Equality requires no decrease of relative entropy on the interval (for a primitive semigroup, equilibrium is the stationary equality case).

Proof. Here $W = 0$. By (6) and Lemma 12, $\beta\Delta F = \Delta D_{\mathrm{KL}} \leq 0$, hence $\mathcal{S} = \Delta F \leq 0$. □

Remark 14 (Driven protocols require a different theorem)**.** *For time-dependent $H_S(t)$ one may define the instantaneous Gibbs state of the full driven Hamiltonian and assume a generator satisfying instantaneous detailed balance, together with inclusive work $\int \operatorname{tr}\Big(\rho\,\dot{H}_S\Big)dt$. Those hypotheses describe a different driven-thermodynamics result and are not proved here. Open-loop alone does not repair the missing relation between generator and reference state; feedback protocols require their own information balance.*

Remark 15 (Memorylessness and temporal correlations)**.** *Under the Born–Markov approximation, the bath correlation function is replaced by its white-noise limit $C(t,s) \to C_0\,\delta(t-s)$, and the GKSL dissipator depends only on the spectral density $J(\omega)$ evaluated at the system's Bohr frequencies. The reduced semigroup therefore omits explicit inter-epoch bath correlations. This modelling loss does not imply a universal spectral gap formula, a universal maintenance-work bound, or impossibility of continuously driven Markovian persistence.*

Remark 16 (Dissipative vs. Self-Nourishing Structures)**.** *The restricted theorem is a free-energy monotonicity statement for one autonomous detailed-balance class. It does not say that Markovian systems cannot persist under continuous driving, that non-Markovianity is necessary for persistence, or that a memory kernel alone supplies usable feedback. It is a controlled baseline for comparisons at fixed initial resources.*

4 Initial-Correlation Accounting

We now separate a dynamics-agnostic accounting identity from the model-dependent question of memory. Quantum mutual information $I(S{:}E)$ can be a consumable initial resource. A non-Markovian model may preserve such a resource across epochs, but non-Markovianity alone does not fix the sign of $\mathcal{S}$ and is neither necessary nor sufficient for a positive interval.

4.1 System–Environment Mutual Information

We work with the total system–environment state $\rho_{SE}(t)$, evolving unitarily under the total Hamiltonian (1). The quantum mutual information

$$I(S{:}E;\, t) := S(\rho_S(t)) + S(\rho_E(t)) - S(\rho_{SE}(t)) = D_{\mathrm{KL}}\big(\rho_{SE}(t) \,\big\|\, \rho_S(t) \otimes \rho_E(t)\big) \geq 0 \tag{14}$$

quantifies the total correlations (classical and quantum) between the system S and the environment E at time t.

Remark 17 (Role of Initial Correlations)**.** *Under the Born approximation, the initial state is taken as a product $\rho_{SE}(0) = \rho_S(0) \otimes \rho_E^{\mathrm{th}}$, so $I(S{:}E;\, 0) = 0$. For a system already interacting with its environment, a restart state* may *be correlated. If a specified memory mechanism preserves those correlations and a later interval consumes them, they contribute to that interval's resource balance. This is a conditional mechanism, not a generic property of every persistent system or memory kernel.*

4.2 The Information–Thermodynamic Identity

The following identity is the central technical tool of this section. It holds for **any** initial state—product or correlated—and relies only on unitarity and the definitions of mutual information and relative entropy. It is the Esposito–Lindenberg–Van den Broeck entropy-production decomposition [14], written so that arbitrary initial-correlation terms are retained explicitly.

Remark 18 (Relative-entropy chain rule)**.** *We repeatedly use the identity*

$$D_{\mathrm{KL}}\big(\rho_{SE} \,\big\|\, \rho_S \otimes \sigma_E\big) = I(S{:}E)_{\rho_{SE}} + D_{\mathrm{KL}}(\rho_E \| \sigma_E), \tag{15}$$

valid for arbitrary *(possibly correlated) ρ_{SE} and any full-rank reference state σ_E.*[1] *Importantly, this is a* pure algebraic identity *and does not assume product initial conditions.*

Lemma 19 (Information–Thermodynamic Identity)**.** *Let $\rho_{SE}(t)$ evolve unitarily under the total Hamiltonian. Then, for* ***any*** *initial state $\rho_{SE}(0)$ (product or correlated):*

$$\Delta I(S{:}E) + \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big) = \Delta S_S + \beta\, \Delta\langle H_E\rangle, \tag{16}$$

where $\Delta S_S = S(\rho_S(\tau)) - S(\rho_S(0))$ is the change in the system's von Neumann entropy and $\Delta\langle H_E\rangle = \mathrm{Tr}[\rho_E(\tau)\, H_E] - \mathrm{Tr}[\rho_E(0)\, H_E]$ is the energy absorbed by the environment.

Proof. Applying the chain rule (15) (Remark 18) with $\sigma_E = \rho_E^{\mathrm{th}}$:

$$D_{\mathrm{KL}}\big(\rho_{SE}(t) \,\big\|\, \rho_S(t) \otimes \rho_E^{\mathrm{th}}\big) = I(S{:}E;\, t) + D_{\mathrm{KL}}\big(\rho_E(t) \,\big\|\, \rho_E^{\mathrm{th}}\big). \tag{17}$$

[1] This follows from the definition of quantum relative entropy and $\ln(\rho_S \otimes \sigma_E) = \ln \rho_S \otimes \mathbb{1}_E + \mathbb{1}_S \otimes \ln \sigma_E$. See, e.g., M. M. Wilde, *Quantum Information Theory*, 2nd ed., Cambridge University Press (2017), Sec. 11; and M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press (2000), Ch. 11.

Expanding the left side using $\ln \rho_E^{\mathrm{th}} = -\beta H_E - \ln Z_E$:

$$D_{\mathrm{KL}}\big(\rho_{SE}(t) \,\big\|\, \rho_S(t) \otimes \rho_E^{\mathrm{th}}\big) = -S(\rho_{SE}(t)) + S(\rho_S(t)) + \beta\langle H_E\rangle_t + \ln Z_E. \tag{18}$$

Since the total evolution is unitary, $S(\rho_{SE}(t)) = S(\rho_{SE}(0))$ for all t. Taking the difference between times τ and 0 cancels both $S(\rho_{SE})$ and $\ln Z_E$, yielding

$$\Delta\big[I(S{:}E) + D_{\mathrm{KL}}(\rho_E \| \rho_E^{\mathrm{th}})\big] = \Delta S_S + \beta\,\Delta\langle H_E\rangle. \tag{19}$$

□

Remark 20 (No assumption on the initial state)**.** *The proof of Lemma 19 uses* only *unitarity ($\Delta S(\rho_{SE}) = 0$) and the algebraic structure of the KL divergence. No assumption is made about the initial state $\rho_{SE}(0)$, the coupling strength, or the character (Markovian or non-Markovian) of the reduced dynamics. For a product thermal initial state, autonomous driving, and negligible endpoint interaction energy, the associated entropy balance reduces to the Esposito decomposition [14]: $\Sigma = I(S{:}E;\,\tau) + D_{\mathrm{KL}}(\rho_E(\tau)\|\rho_E^{\mathrm{th}}) + O(\lambda)$.*

4.3 The Survival Identity

We now connect the information–thermodynamic identity (16) to the survival functional $\mathcal{S}$ defined in Section 2.4.

Theorem 21 (Survival Balance: General Form)**.** *Under Assumptions* (A1)–(A5) *of Definition 6, let $\rho_{SE}(t)$ evolve unitarily from an* arbitrary *(possibly correlated) initial state $\rho_{SE}(0)$. The survival functional satisfies*

$$\boxed{\beta\,\mathcal{S}[\Lambda,\tau] = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big) - \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle - \beta\,\Delta\langle H_{\mathrm{int}}\rangle,} \tag{20}$$

where $\Delta\langle H_{\mathrm{ctrl}}\rangle = \mathrm{Tr}[\rho_S(\tau)\,H_{\mathrm{ctrl}}(\tau)] - \mathrm{Tr}[\rho_S(0)\,H_{\mathrm{ctrl}}(0)]$ is the change in the control-field energy.

*For **autonomous evolution** with negligible interaction-energy change at the endpoints ($H_{\mathrm{ctrl}} = 0$ and $\Delta\langle H_{\mathrm{int}}\rangle = O(\lambda)$), the weak-coupling form is*

$$\beta\,\mathcal{S}[\Lambda,\tau] = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big) + O(\lambda). \tag{21}$$

Proof. The proof uses three ingredients: the definition of $\mathcal{S}$, the first law, and Lemma 19.
Step 1 (First law in weak coupling). Since $H_{\mathrm{ctrl}}(t)$ is the only explicitly time-dependent component of H, inclusive work satisfies the exact first law $W = \Delta\langle H\rangle = \Delta\langle H_R\rangle + \Delta\langle H_{\mathrm{ctrl}}\rangle + \Delta\langle H_E\rangle + \Delta\langle H_{\mathrm{int}}\rangle$.
Step 2 (Connecting Σ to the identity). From Definition 8 and (8), using $\Delta F = \Delta\langle H_R\rangle - \beta^{-1}\Delta S_S$:

$$\begin{aligned}\Sigma &= \beta(W - \Delta F) = \beta\big(W - \Delta\langle H_R\rangle\big) + \Delta S_S \\ &= \beta\big(\Delta\langle H_{\mathrm{ctrl}}\rangle + \Delta\langle H_E\rangle + \Delta\langle H_{\mathrm{int}}\rangle\big) + \Delta S_S \\ &= \big(\Delta S_S + \beta\,\Delta\langle H_E\rangle\big) + \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle + \beta\,\Delta\langle H_{\mathrm{int}}\rangle.\end{aligned} \tag{22}$$

By Lemma 19, the parenthesized term equals $\Delta I(S{:}E) + \Delta D_{\mathrm{KL}}(\rho_E \| \rho_E^{\mathrm{th}})$. Hence

$$\Sigma = \Delta I(S{:}E) + \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big) + \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle + \beta\,\Delta\langle H_{\mathrm{int}}\rangle. \tag{23}$$

Step 3 (Survival functional). $\beta\,\mathcal{S} = -\Sigma$ by (10), yielding (20). For $H_{\mathrm{ctrl}} = 0$ and weak endpoint coupling, the last term is $O(\lambda)$, yielding (21). □

Remark 22 (Nature of the result)**.** *Equation* (20) *is an exact* accounting identity*, not an inequality or optimality bound. It establishes that the sign is governed by the declared initial resources and endpoint bookkeeping: changes in correlation, bath displacement, control energy, and interaction energy. The identity itself is dynamics-agnostic and does not make non-Markovianity a resource. A memory kernel matters only when a model shows that it preserves correlations across epochs and later permits their consumption.*

Remark 23 (Scope of the theorem)**.** *Theorem 21 holds for* any *initial state $\rho_{SE}(0)$—product or correlated. The proof requires only Assumptions* (A1)–(A5) *of Definition 6 and the definitions of $\mathcal{S}$, $I(S{:}E)$, and D_{KL}. No assumption about the reduced dynamics (Markovian, non-Markovian, or otherwise) is needed. An interacting system may carry correlations at the chosen interval boundary. When present and subsequently consumed, those correlations are one thermodynamic resource; persistence does not require that every model use this resource.*

4.4 Three Regimes of Survival

We specialize to the autonomous, endpoint-decoupled weak-coupling convention ($H_{\mathrm{ctrl}} = 0$ and the $O(\lambda)$ interaction term suppressed). This isolates the initial-resource accounting.

Corollary 24 (Three Regimes)**.** *Under autonomous evolution, identity* (21) *identifies three regimes:*

1. ***Product initial state*** *($I(S{:}E;0) = 0$, $D_{\mathrm{KL}}(\rho_E(0)\|\rho_E^{\mathrm{th}}) = 0$): Both ΔI and ΔD_{KL} are increases from zero to non-negative final values, so*

$$\beta\,\mathcal{S} = -\big(I(S{:}E;\tau) + D_{\mathrm{KL}}(\rho_E(\tau)\|\rho_E^{\mathrm{th}})\big) \le 0.$$

   This product-state unitary balance is dynamics-agnostic and is compatible with, but logically distinct from, the restricted Davies result (Theorem 13).

2. ***Correlated initial state*** *($I(S{:}E;0) > 0$): If the dynamics* consumes *pre-existing correlations ($\Delta I < 0$, i.e., $I(S{:}E;\tau) < I(S{:}E;0)$), the first term contributes* positively *to $\mathcal{S}$. Provided*

$$|\Delta I(S{:}E)| > \Delta D_{\mathrm{KL}}\big(\rho_E \,\big\|\, \rho_E^{\mathrm{th}}\big), \tag{24}$$

   the survival functional is strictly positive: $\mathcal{S} > 0$. The agent has converted pre-existing correlations into usable free energy.

3. ***Upper bound:*** *Since $I(S{:}E;\tau) \ge 0$ and $D_{\mathrm{KL}}(\rho_E(\tau)\|\rho_E^{\mathrm{th}}) \ge 0$, the maximum survival gain is bounded by*

$$\beta\,\mathcal{S} \le I(S{:}E;0) + D_{\mathrm{KL}}\big(\rho_E(0) \,\big\|\, \rho_E^{\mathrm{th}}\big). \tag{25}$$

   The thermodynamic profit cannot exceed the total initial "resource budget"—the pre-existing correlations plus the initial displacement of the bath from equilibrium.

4.5 The Correlation Battery

The three regimes of Corollary 24 raise a natural question: *where do the initial correlations* $I(S{:}E;\,0) > 0$ *come from?*

Remark 25 (The Correlation Battery)**.** *The answer is: from* ***prior interaction epochs****. A persistent agent does not begin its existence in a product state. Over any interaction interval, unitary evolution generically builds system–environment correlations (*$\Delta I > 0$*), at a thermodynamic cost (*$\mathcal{S} < 0$ *during this phase by Corollary 24(i)). A non-Markovian dilation can preserve correlations across the chosen epoch boundary, but whether they are later consumed with* $\mathcal{S} > 0$ *must be checked from the full balance.*

The process is analogous to a ***battery****:*

- ***Charging phase*** *(correlation building,* $\Delta I > 0$*): the agent "pays" free energy to build system–environment correlations.* $\mathcal{S} < 0$*.*
- ***Discharging phase*** *(correlation consumption,* $\Delta I < 0$*): the agent extracts free energy from the stored correlations.* $\mathcal{S} > 0$*.*

The repeated-product Born approximation removes this resource by construction at each reset. A reduced semigroup law by itself does not state that every microscopic dilation is uncorrelated. Conversely, a memory kernel only represents retained history; it does not guarantee that the history is usable as thermodynamic work.

Crucially, global thermodynamics remains respected. For any full cycle starting from an uncorrelated thermal state ($I(S{:}E;\,0) = 0$*,* $D_{\mathrm{KL}}(\rho_E(0)\|\rho_E^{\mathrm{th}}) = 0$*), the total survival functional satisfies*

$$\beta\,\mathcal{S}[0,t] = -\Sigma[0,t] \;\le\; 0 \qquad \textit{(second law)}. \tag{26}$$

The local positivity $\mathcal{S}[t^*,t] > 0$ *during the discharging phase is strictly funded by the free energy dissipated during the earlier charging phase (see Proposition 26 for the formal decomposition).*

Proposition 26 (Full-cycle closure)**.** *Under the conditions of Theorem* 21 *with autonomous evolution (*$H_{\mathrm{ctrl}} = 0$*) and negligible interaction-energy changes at the endpoints of each displayed balance, partition* $[0,\tau]$ *at any intermediate time* t^* *into a charging phase* $[0,t^*]$ *and a discharging phase* $[t^*,\tau]$*.*

(i) ***Charging*** *(product initial state,* $I(S{:}E;\,0) = 0$*). By Corollary* 24(i)*,*

$$\beta\,\mathcal{S}[0,t^*] = -I(S{:}E;\,t^*) - D_{\mathrm{KL}}\big(\rho_E(t^*)\,\big\|\,\rho_E^{\mathrm{th}}\big) \;\le\; 0. \tag{27}$$

(ii) ***Discharging*** *(correlated initial state at* t^**). Applying* (21) *to* $[t^*,\tau]$*:*

$$\beta\,\mathcal{S}[t^*,\tau] = -\big(I(S{:}E;\,\tau) - I(S{:}E;\,t^*)\big) - \big(D_{\mathrm{KL}}(\rho_E(\tau)\|\rho_E^{\mathrm{th}}) - D_{\mathrm{KL}}(\rho_E(t^*)\|\rho_E^{\mathrm{th}})\big), \tag{28}$$

which is positive whenever the decrease in correlations dominates the change in bath displacement (Corollary 24(ii)*).*

(iii) ***Full cycle.*** *Since $\mathcal{S}$ is additive over concatenated intervals, $\beta\,\mathcal{S}[0,\tau] = \beta\,\mathcal{S}[0,t^*] + \beta\,\mathcal{S}[t^*,\tau]$. Equivalently, applying* (21) *directly to $[0,\tau]$ with $I(S{:}E;\,0) = 0$:*

$$\beta\,\mathcal{S}[0,\tau] = -I(S{:}E;\,\tau) - D_{\mathrm{KL}}\big(\rho_E(\tau)\,\big\|\,\rho_E^{\mathrm{th}}\big) \;\le\; 0. \tag{29}$$

The net thermodynamic profit over the full cycle is non-positive—the "interest" paid during charging meets or exceeds the "dividend" collected during discharging. But the local *positivity of $\mathcal{S}$ during discharge* (28) *is what enables the agent to survive through intervals that would kill a memoryless system.*

Proof. The survival functional is additive over concatenated intervals:

$$\mathcal{S}[0,\tau] = \underbrace{\big(\Delta F[0,t^*] - W[0,t^*]\big)}_{\mathcal{S}[0,t^*]} + \underbrace{\big(\Delta F[t^*,\tau] - W[t^*,\tau]\big)}_{\mathcal{S}[t^*,\tau]},$$

since both ΔF and W decompose additively. Items (i) and (iii) then follow from Theorem 21 (autonomous case) applied to $[0,t^*]$ and $[0,\tau]$ respectively, each starting from a product state. Item (ii) follows from Theorem 21 applied to $[t^*,\tau]$ with correlated initial state $\rho_{SE}(t^*)$. Inequality (29) holds because $I(S{:}E;\,\tau) \ge 0$ and $D_{\mathrm{KL}}(\rho_E(\tau)\|\rho_E^{\mathrm{th}}) \ge 0$. □

4.6 Connection to Non-Markovianity Measures

Remark 27 (The Born Approximation Destroys the Resource)**.** *Under the Born (product-state) approximation, every infinitesimal time step begins from $\rho_{SE} \approx \rho_S \otimes \rho_E^{\mathrm{th}}$, enforcing $I(S{:}E) = 0$ at all times. Corollary 24(i) then guarantees $\mathcal{S} \le 0$ for* every *finite interval. The Born approximation does not merely simplify the dynamics—it* eliminates the thermodynamic resource *(system–environment correlations) that would otherwise be available.*

Remark 28 (Connection to BLP Non-Markovianity)**.** *The Breuer–Laine–Piilo (BLP) measure of non-Markovianity [13] is defined via the temporary increase of trace distance between pairs of initial states: $\mathcal{N}_{\mathrm{BLP}} := \max_{\rho_{1,2}} \int_{\dot D>0} \frac{d}{dt} D(\rho_1(t),\rho_2(t))\,dt$. BLP backflow and correlation consumption can coincide in particular models, but neither condition implies the other in general [9]. The BLP measure therefore diagnoses one reduced-dynamics notion of non-Markovianity; it is not a witness for the sign of $\mathcal{S}$ without the accompanying energy and correlation balance.*

Remark 29 (Consistency with the Sagawa–Ueda Framework)**.** *In the Sagawa–Ueda framework [7, 8], measurement-based feedback permits $\Sigma \ge -I_{\mathrm{feedback}}$, where I_{feedback} is the mutual information gained through measurement. Pre-existing correlations are an analogous accounting resource, but a memory kernel is not itself a measurement record and the two protocols must not be identified. The total system (agent + bath) still satisfies $\Sigma_{\mathrm{total}} \ge 0$; the apparent "profit" for the agent is paid for by the correlations consumed from the system–environment entanglement. The bound* (25) *is the non-Markovian analogue of the Sagawa–Ueda bound $\beta\,\mathcal{S} \le I_{\mathrm{feedback}}$.*

4.7 Mechanism: The Surfer Analogy

The physical mechanism admits an intuitive picture.

- **The Markovian Agent (The Stone):** A stone thrown into the ocean sinks. It interacts with the water only at the instant of contact, dissipates its kinetic energy, and thermalizes ($\mathcal{S} \leq 0$). Each collision builds system–environment correlations that are immediately discarded (Born approximation), so $I(S{:}E) = 0$ at all times. The wave structure is invisible to it.
- **The Non-Markovian Agent (The Surfer):** A surfer carries *memory* of past wave patterns—encoded in the correlations $I(S{:}E;\, t_0) > 0$ built up over previous interactions (the "charging phase" of Remark 25). During backflow intervals ($\Delta I < 0$), the surfer *spends* these stored correlations to extract free energy from the wave itself. The surfer remains far from equilibrium not by fighting the environment, but by converting temporal correlations into thermodynamic profit.

Remark 30 (Thermodynamic Rectification)**.** *The "surfing" mechanism is* ***thermodynamic rectification****: the memory kernel $\mathcal{K}(t, s)$ functions as a temporal filter that enables the system to accumulate correlations during one phase and consume them during another. Formally, the kernel enables access to the resource $I(S{:}E;\, 0)$ accumulated during previous interaction epochs—converting the environment's temporal correlations into the system's structural persistence via the ΔI term in Theorem 21.*

Remark 31 (Memory as Implicit Maxwell's Demon)**.** *The memory kernel functions as an* implicit Maxwell's demon*. A Markovian system interacts with each environmental fluctuation exactly once, at the moment of contact; the Born approximation resets $I(S{:}E) = 0$ after each step. A non-Markovian system retains a trace of past fluctuations (via $\mathcal{K}(t, s)$ with $s < t$) and can exploit correlations between past and present environmental states. This is not a violation of the second law but an instance of the Sagawa–Ueda generalization: the demon's cost is paid in the currency of memory maintenance (Landauer erasure), a point we quantify in Section 7. The total budget for "demonic profit" is capped by the bound* (25).

5 Operational Temporal Relations

The accounting identity does not derive a temporal order. This section records two heuristic structures that can be studied after a reduced memory model is specified: an operational influence relation between times and a comparison with pointer-record growth. Neither is a universal partial order, and no HAFF theorem is inherited.

5.1 Thresholded Memory-Influence Relation

A non-Markovian memory kernel $\mathcal{K}(t, s)$ defines a causal link between a past state at s and the present dynamics at t. We define a thresholded relation based on the effective response.

Definition 32 (Operational memory-influence relation)**.** *For a declared reduced model, let $\mathcal{R}_{t\leftarrow s}$ denote the linear response of the predicted state or read-out at time t to a perturbation applied at time $s < t$. At tolerance $\epsilon > 0$, define*

$$s \rightsquigarrow_K t \quad \Longleftrightarrow \quad \|\mathcal{R}_{t\leftarrow s}\| > \epsilon. \tag{30}$$

The response norm, perturbation class, and tolerance are part of the protocol. This relation need not be transitive or antisymmetric, so it is not asserted to be a partial order.

An ideal delta-memory reduced model has no off-diagonal response in this description. A finite-memory model can have $s \rightsquigarrow_K t$ within its response window. This statement is about the chosen reduced model, not about the existence of time or a universal thermodynamic arrow.

5.2 A Fisher-Accretion Conjecture

One may ask whether a specified survival optimizer tends to retain parameter information in the forward protocol direction. The following statement is a conjecture; the survival identity alone does not create statistical irreversibility.

Conjecture 33 (Fisher Information Accretion)**.** *Let $\mathcal{I}_F(\theta;\, \rho(t))$ denote the Fisher information contained in the system state $\rho(t)$ regarding a parameter θ encoded in the environment at time $s < t$. For an agent whose dynamics maximize the survival functional* (9)*, the time-averaged Fisher information satisfies*

$$\overline{\frac{d}{dt}\,\mathcal{I}_F(\theta;\, \rho(t))} \geq 0, \tag{31}$$

where the overbar denotes a time average over scales larger than the bath correlation time τ_B.

Remark 34 (Status of the conjecture)**.** *The survival balance supplies no optimization domain, controller class, or monotonicity theorem for Fisher information. Mutual information and Fisher information are distinct functionals, and information backflow can make either non-monotone. Establishing the displayed inequality requires a concrete controlled process and objective; it is not used as an input to later theorems.*

5.3 The Bridge to HAFF

Revised HAFF Paper F [22] distinguishes an algebra-inclusion preorder from a protocol-relative redundancy diagnostic and does not prove generic monotone record expansion. The following comparison is therefore a heuristic interface, not a theorem.

Conjecture 35 (Conditional pointer-record comparison)**.** *Suppose a concrete branching/record model supplies (i) a pointer coarse-graining Φ, (ii) a fixed fragmentation, and (iii) a monotone record channel for a specified observation window. One may then test whether an influence $s \rightsquigarrow_K t$ is accompanied by nested record algebras,*

$$\mathcal{R}_s \subseteq \mathcal{R}_t. \tag{32}$$

The memory-influence relation alone does not imply this inclusion; backflow, record erasure, or a changed fragmentation can violate it.

Remark 36 (Scope of the Correspondence)**.** *Conjecture 35 states a test that can succeed in a branching record model. No general monotonicity, embedding, or duality between T-DOME and HAFF is established.*

Remark 37 (Dynamical and Algebraic Time)**.** *The two papers use distinct, protocol-dependent notions:*

	Paper F (HAFF)	***Paper I (T-DOME)***
Nature	*Algebraic*	*Dynamical*
Mechanism	*Record/algebra diagnostics*	*Reduced-model response*
Formalism	*Preorders/diagnostics*	*Relation $\rightsquigarrow_K$ on times*
Asymmetry source	*Declared record protocol*	*Declared kernel/response model*
Domain	*Classical (pointer) sector*	*Full quantum dynamics*

An explicit morphism between them remains open.

Remark 38 (The Seed and the Tree)**.** *The seed/tree language is a programme metaphor. It supplies neither a time variable nor a theorem connecting memory dynamics to geometry.*

6 Worked Example: The Quantum Predictive Agent

To illustrate free-energy monotonicity and a correlation-funded backflow interval *within one explicit model*, we employ an open quantum system model: the spin-boson model with Lorentz–Drude spectral density, which admits an exact analytic solution for the decoherence dynamics [1].

6.1 Model Setup

The total Hamiltonian is $H = H_S + H_B + H_I$. The agent is a two-level system with energy gap ω_0: $H_S = \frac{\omega_0}{2}\sigma_z$. The environment is a bosonic bath: $H_B = \sum_k \omega_k b_k^\dagger b_k$. The interaction is of the pure-dephasing form $H_I = \sigma_z \otimes \sum_k (g_k\, b_k + g_k^*\, b_k^\dagger)$.

The spectral density $J(\omega) = \sum_k |g_k|^2 \delta(\omega - \omega_k)$ characterizes the environment. We choose the Lorentz–Drude form:

$$J(\omega) = \frac{2\lambda\,\gamma\,\omega}{\omega^2 + \gamma^2}, \tag{33}$$

where λ is the reorganization energy and γ is the bath memory rate (inverse correlation time $\tau_B = 1/\gamma$). We place the system in the low-temperature regime $\beta\omega_0 \gg 1$ (i.e., $k_B T \ll \omega_0$). The bath correlation function in the $T \to 0$ limit is $C(t) = \lambda\gamma\, e^{-\gamma|t|}$, so the parameter γ directly controls the bath memory depth. For $\beta\omega_0 \geq 10$ the finite-temperature corrections to all quantities below are of order $O(e^{-\beta\omega_0}) \lesssim 5 \times 10^{-5}$ and are neglected throughout.[2]

[2]All plots and numerical values use the standard $T \to 0$ analytic expression for the decoherence function (34) (see Breuer and Petruccione [1], Sec. 12.3, for the Lorentz–Drude pure-dephasing solution), which provides an accurate proxy in the low-temperature regime; β is a well-defined bookkeeping parameter and β^{-1} a finite energy scale.

6.2 Exact Decoherence Function

For the pure-dephasing spin-boson model in the $T \to 0$ limit, the off-diagonal element of the reduced density matrix $\rho_{01}(t) = \rho_{01}(0)\, p(t)$ is governed by the **decoherence function** [1]:

$$p(t) = e^{-\gamma t/2} \left[\cos(\Omega t) + \frac{\gamma}{2\Omega} \sin(\Omega t) \right], \tag{34}$$

where $\Omega := \frac{1}{2}\sqrt{4\lambda\gamma - \gamma^2}$. This solution is exact for the Lorentz–Drude spectral density.

Remark 39 (Non-Markovian Regime)**.** *The character of the dynamics is controlled by the discriminant* $\Delta := 4\lambda\gamma - \gamma^2 = \gamma(4\lambda - \gamma)$*:*

- $\gamma > 4\lambda$ *(*$\Delta < 0$*):* Ω *is imaginary,* $p(t)$ *decays monotonically. The dynamics is Markovian (no backflow).*
- $\gamma = 4\lambda$ *(*$\Delta = 0$*): Critical damping.* $p(t) = (1 + \gamma t/2)\, e^{-\gamma t/2}$*.*
- $\gamma < 4\lambda$ *(*$\Delta > 0$*):* Ω *is real and positive.* $p(t)$ *oscillates with envelope* $e^{-\gamma t/2}$*. The dynamics is* non-Markovian*: intervals where* $|p(t)|$ *increases correspond to information backflow [13].*

The non-Markovian regime $\gamma < 4\lambda$ *is thus the regime of structured, long-memory baths.*

6.3 Quantitative Evaluation

We now evaluate the survival functional explicitly. For the pure-dephasing model, populations are conserved ($p_0(t) = p_0(0)$, $p_1(t) = p_1(0)$), and the non-equilibrium free energy depends only on the coherence:

$$F(\rho(t)) - F(\rho_{\mathrm{eq}}) = \beta^{-1}\, D_{\mathrm{KL}}(\rho(t) \| \rho_{\mathrm{eq}}). \tag{35}$$

For a qubit with initial state $\rho(0) = \frac{1}{2}(\mathbb{1} + \vec{r} \cdot \vec{\sigma})$ and $r_z = 0$ (maximal coherence in the x–y plane), the relative entropy reduces to $D_{\mathrm{KL}}(\rho(t)\|\rho_{\mathrm{eq}}) \approx |p(t)|^2\, |\rho_{01}(0)|^2$ to leading order in the coherence (see, e.g., [1]). Since there is no external driving ($H_{\mathrm{ctrl}} = 0$, $W = 0$), the survival functional is simply

$$\beta\, \mathcal{S}(t) = D_{\mathrm{KL}}(\rho(t)\|\rho_{\mathrm{eq}}) - D_{\mathrm{KL}}(\rho(0)\|\rho_{\mathrm{eq}}) \propto |p(t)|^2 - 1. \tag{36}$$

The proportionality in (36) is specific to the **pure-dephasing model** with the chosen maximally coherent initial state ($r_z = 0$) and measurement in the pointer basis (σ_z). Under these conditions, the exact solution [1] ensures that population terms vanish from the free energy ($\Delta\langle H_S \rangle = 0$), leaving only the coherence contribution: $\beta\, \mathcal{S} = -\Delta S_S$ depends only on the coherence trajectory $|p(t)|$. The proxy $|p(t)|^2$ thus rigorously captures the sign and monotone behaviour of $\beta\, \mathcal{S}$; the exact numerical prefactor depends on the initial state and on β, but the qualitative conclusion—$\mathcal{S} > 0$ during backflow intervals—is robust and does not depend on the proxy normalization.

For a Markovian evolution, $|p(t)|$ decreases monotonically, so $|p(t)|^2 - 1 \leq 0$ for all t: $\mathcal{S} \leq 0$ always in this autonomous model (consistent with Theorem 13). For non-Markovian

evolution with $\gamma < 4\lambda$, the oscillations in $p(t)$ produce intervals where $|p(t)|$ *increases after a previous decrease*, i.e., the system *re-coheres*.

Concrete parameters. Set $\omega_0 = 1$ (energy units), $\lambda = 1$, $\gamma = 0.5$ (deep non-Markovian regime: $\gamma/4\lambda = 0.125 \ll 1$). Then:

$$\Omega = \tfrac{1}{2}\sqrt{4 \cdot 1 \cdot 0.5 - 0.25} = \tfrac{1}{2}\sqrt{1.75} \approx 0.661. \tag{37}$$

The decoherence function (34) first reaches $p(t^*) = 0$ at $t^* = [\pi/2 + \arctan\big(\gamma/(2\Omega)\big)]/\Omega \approx 1.93/\Omega \approx 2.92$ (in units of ω_0^{-1}), where the system has fully decohered. Subsequently, the environment *returns* coherence: $|p(t)|$ increases, reaching a local maximum $|p(t_1)| \approx 0.31$ at $t_1 \approx 4.75/\omega_0$.

Over the backflow interval $[t^*, t_1]$, for the pure-dephasing qubit with the chosen initial state ($r_z = 0$, maximal coherence) and in the autonomous setting ($H_{\rm ctrl} = 0$, $W = 0$), the survival proxy (36) gives

$$\beta\,\mathcal{S}[t^*, t_1] \propto |p(t_1)|^2 - |p(t^*)|^2 \approx 0.093 - 0 = 0.093 > 0. \tag{38}$$

Equivalently, $\mathcal{S} \approx 0.093\,\beta^{-1}$ in the bookkeeping units set by β. The agent has gained a dimensionless survival advantage $\beta\,\mathcal{S} \approx +0.093$ *with zero work input* (autonomous evolution, $H_{\rm ctrl} = 0$), solely by exploiting the non-Markovian backflow. Figure 1 illustrates the contrast between Markovian and non-Markovian evolution.

Consistency with Theorem 21 and Proposition 26. Since this is autonomous evolution ($H_{\rm ctrl} = 0$), identity (21) applies exactly: $\beta\,\mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\rm KL}(\rho_E \| \rho_E^{\rm th})$. The example realizes the *correlation battery* of Remark 25, with the charge–discharge decomposition of Proposition 26:

- **Charging** ($[0, t^*]$, eq. (27)): the system decoheres, building correlations $I(S{:}E;\, t^*) > 0$ at the cost of $\mathcal{S} < 0$.
- **Discharging** ($[t^*, t_1]$, eq. (28)): the correlations are consumed ($\Delta I < 0$ over this interval), returning $\beta\,\mathcal{S} \approx +0.093 > 0$.

The bound (25) is satisfied: $\beta\,\mathcal{S}[t^*, t_1] = 0.093 \le I(S{:}E;\, t^*)$. Full-cycle closure (29) is confirmed: $\beta\,\mathcal{S}[0, t_1] < 0$.

Instantaneous decoherence rate. The rate of coherence loss is

$$\Gamma(t) := -\frac{d}{dt}\ln|p(t)| = \frac{\gamma}{2} - \frac{\Omega\sin(\Omega t) + \frac{\gamma}{2}\cos(\Omega t)}{\cos(\Omega t) + \frac{\gamma}{2\Omega}\sin(\Omega t)}\,. \tag{39}$$

In the Markovian limit $\gamma \gg 4\lambda$, $\Gamma(t) \to \gamma/2 > 0$ for all t (monotone decoherence). In the non-Markovian regime $\gamma < 4\lambda$, $\Gamma(t)$ oscillates and becomes *negative* during the backflow intervals where $|p(t)|$ increases. These are precisely the intervals where $\mathcal{S} > 0$.

Closed-form revival amplitude. The decoherence function (34) can be written as $p(t) = R\,e^{-\gamma t/2}\cos(\Omega t - \phi)$, where $R = \sqrt{1 + (\gamma/2\Omega)^2}$ and $\phi = \arctan(\gamma/2\Omega)$, with $R\cos\phi = 1$. The extrema of $|p(t)|$ occur at $t_n = n\pi/\Omega$ ($n = 0, 1, 2, \ldots$), and the first revival peak after the first zero is at $t_1 = \pi/\Omega$. Its amplitude is *exactly*

$$|p(t_1)| = e^{-\gamma\pi/(2\Omega)}, \qquad \beta\,\mathcal{S}[t^*, t_1] \approx |p(t_1)|^2 = e^{-\gamma\pi/\Omega}. \tag{40}$$

This is the paper's central computable prediction: the survival gain at first backflow is determined by a single dimensionless ratio γ/Ω.

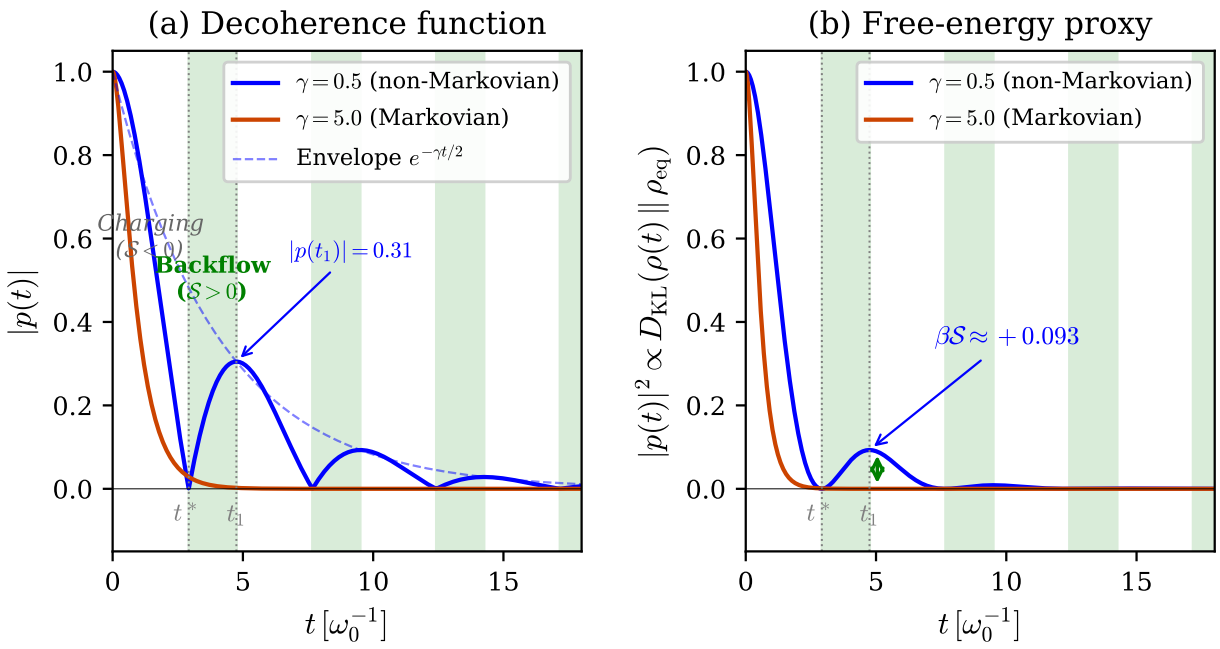


Figure 1: Pure-dephasing spin-boson model (Section 6) with Lorentz–Drude spectral density (33). **Parameters:** $\omega_0 = 1$ (energy unit), $\lambda = 1$ (reorganization energy). **Units:** all times in ω_0^{-1}; energies in $\hbar\omega_0$. **Regime:** low temperature ($\beta\omega_0 \gg 1$); the standard $T \to 0$ analytic expression (34) [1] is used as an accurate proxy. **(a)** Decoherence amplitude $|p(t)|$ (eq. (34)). Blue: non-Markovian ($\gamma = 0.5$, $\gamma/4\lambda = 0.125$). Orange: Markovian ($\gamma = 5.0$, $\gamma/4\lambda = 1.25$). Dashed: exponential envelope $e^{-\gamma t/2}$. Green bands indicate backflow (Remark 39: $d|p|/dt > 0$, $\Gamma(t) < 0$ per (39)). **(b)** Survival proxy $|p(t)|^2 \propto \beta\,\mathcal{S}$ (eq. (36)). At the first revival ($t_1 = \pi/\Omega \approx 4.75\,\omega_0^{-1}$), the non-Markovian agent achieves $\beta\,\mathcal{S}[t^*, t_1] \approx +0.093$ (eq. (38)), consistent with the closed-form prediction (40), funded by the consumption of pre-existing correlations (Proposition 26). The Markovian comparison decays monotonically: $\mathcal{S} \leq 0$ in this autonomous model (Theorem 13).

Remark 40 (Parameter Survey)**.** *Table 1 demonstrates the transition from the Markovian regime ($\mathcal{S} \leq 0$) to the non-Markovian regime ($\mathcal{S} > 0$) as the bath memory rate γ decreases below the critical value 4λ. All entries use $\omega_0 = 1$, $\lambda = 1$, $W = 0$ (autonomous evolution), with revival amplitudes computed from* (40)*.*

γ	$\gamma/4\lambda$	*Regime*	$\lvert p(t_1)\rvert$	$\beta\,\mathcal{S}(t_1)$	$\Gamma_{\min}$
20.0	5.0	*Markov*	—	≤ 0	> 0
4.0	1.0	*Critical*	—	≤ 0	$= 0$
2.0	0.50	*Non-Markov*	0.043	+0.002	< 0
1.0	0.25	*Non-Markov*	0.163	+0.027	< 0
0.5	0.125	*Deep NM*	0.305	+0.093	< 0
0.1	0.025	*Deep NM*	0.605	+0.37	< 0

Table 1: Survival functional at first backflow revival as a function of the bath memory rate γ, for the pure-dephasing spin-boson model with Lorentz–Drude spectral density. $|p(t_1)|$ is computed from (40); $\Gamma_{\min}$ is the sign of the minimum of the instantaneous decoherence rate (39). The transition $\mathcal{S} \leq 0 \to \mathcal{S} > 0$ occurs precisely at the non-Markovian threshold $\gamma = 4\lambda$. For $\gamma = 0.1$ (deep non-Markovian), the agent achieves $\beta\,\mathcal{S} \approx +0.37$ per backflow cycle in the autonomous setting ($H_{\text{ctrl}} = 0$).

Remark 41 (The Two Regimes: Summary)**.**

	Markovian ($\gamma = 20$)	***Non-Markovian*** ($\gamma = 0.5$)
$\gamma/4\lambda$	5.0 *(overdamped)*	0.125 *(underdamped)*
τ_B	$0.05\,\omega_0^{-1}$	$2.0\,\omega_0^{-1}$
$p(t)$	*Monotone decay*	*Oscillatory with envelope*
$\lvert p(t_1)\rvert$ *at first revival*	0 *(no revival)*	≈ 0.31
$\beta\,\mathcal{S}$ *at revival*	≤ 0	$\approx +0.093$
$\Gamma(t)$	> 0 *always*	*Oscillates,* < 0 *during backflow*
Interpretation	*Stone (sinks)*	*Surfer (rides backflow)*

The non-Markovian agent achieves $\beta\,\mathcal{S} \approx +0.093$ per backflow cycle (autonomous, $H_{\text{ctrl}} = 0$), while the Markovian agent can only lose free energy. As the coupling deepens ($\gamma/4\lambda \to 0$), the revival amplitude grows and $\mathcal{S}$ increases (Table 1), bounded above by $\beta\,\mathcal{S} \leq I(S{:}E;\, t^*)$ (Corollary 24(iii)).

7 The Cost of Memory

The preceding identity allows a correlated initial resource to fund a later interval. It does not assign a universal energetic price to storing or maintaining that resource. This section separates Landauer's erasure bound from a conditional archive-overload model used only to motivate Paper II.

7.1 The Landauer Debt

Landauer's principle prices a logically irreversible reset, not passive storage, sensing, correlation maintenance, or reversible computation.

Remark 42 (Conditional erasure cost)**.** *If a declared memory implementation irreversibly erases N_{erase} unbiased bits into a bath at temperature T, its reset work obeys*

$$W_{\text{reset}} \geq k_B T \ln 2\, N_{\text{erase}}. \tag{41}$$

No lower bound proportional to total stored mutual information follows without an implementation model. Reversible storage and reversible measurement can in principle avoid this reset cost until a logically irreversible operation is performed [11, 12]. The former universal "Landauer Cost of Memory" proposition is withdrawn.

7.2 Raw Archive Overload Scenario

The crisis arises from the scaling of $\mathcal{I}_{\text{stored}}$ with time. To quantify this, we borrow two quantities from computational mechanics [16, 17]:

Definition 43 (Entropy Rate and Predictive Information)**.** *Let $\{X_t\}$ be the stochastic process describing the environment's influence on the agent (e.g., the sequence of bath correlation values).*

1. *The **entropy rate** of the environment is*

$$h_\mu := \lim_{n\to\infty} H(X_n \mid X_{n-1}, \ldots, X_1), \tag{42}$$

   measuring the intrinsic unpredictability per time step.

2. *The **predictive information** (excess entropy) is*

$$I_{\text{pred}} := I(\overleftarrow{X}; \overrightarrow{X}) = \sum_{k=1}^{\infty} \big[H(X_k) - h_\mu\big], \tag{43}$$

   where $\overleftarrow{X}$ and $\overrightarrow{X}$ denote the past and future half-chains. This is the total amount of information about the future that is encoded in the past—the useful *memory.*

Finite predictive information does not by itself prove that a finite physical memory captures every useful feature, and positive entropy rate does not lower-bound predictive information by $h_\mu \tau_{\text{mem}}$. In particular, a stationary mixing process commonly has finite excess entropy.

Remark 44 (Conditional archive-overload scenario)**.** *Consider a deliberately specified implementation: an uncompressed raw archive receives records at rate r_{raw} bits per unit time, has capacity M bits, and evaluates the entire archive once per control cycle. Without deletion it fills after M/r_{raw}. If it irreversibly resets discarded records at rate r_{erase}, Landauer gives*

$$\dot{W}_{\text{reset}} \geq k_B T \ln 2 \, r_{\text{erase}}. \tag{44}$$

For a long-memory or non-stationary source with divergent I_{pred}, the amount of predictively relevant history may also grow, but that is an additional source assumption. This archive can exceed a finite storage, throughput, or reset budget. The result is an engineering overload scenario, not a universal thermodynamic catastrophe and not a theorem that every memory implementation must compress.

7.3 Conditional Motivation for Compression

An agent using the raw-archive implementation may respond by increasing capacity, lowering sample rate, reversible uncomputation, sufficient statistics, lossy compression, or accepting stale data. Paper II studies one coordinate-selection model among these options.

- **Compression requires a criterion.** To decide what to keep and what to erase, the agent needs a *relevance function*—a mapping from stored correlations to survival value. This is a reference frame that ranks information by its contribution to $\mathcal{S}$.
- **Coordinate selection uses a reference frame.** If the admissible compressor is restricted to retaining a subset of supplied coordinates, choosing that subset privileges a basis. More general invariant or learned sufficient-statistic compressors need not have this form.

Remark 45 (The Origin of Paper II)**.** *Remark 44 supplies Paper II with a conditional motivation: a bounded agent attempting to evaluate an uncompressed archive needs some resource-allocation policy. It does not prove that compression, Clifford coordinates, spontaneous symmetry breaking, or a "self" is thermodynamically necessary. Those are modelling choices to be assessed in Paper II.*

8 Numerical Demonstration

The preceding sections establish an autonomous thermodynamic baseline and an accounting identity. The following HMM is a separate classical performance illustration: it compares finite observation windows with a Bayes filter in an aliased environment. Reward here is not the thermodynamic survival functional, so the demonstration does not test either theorem. Full code and parameters are provided for reproducibility.

8.1 Model

Environment. A two-hidden-state HMM with aliased observations. The hidden state $s_t \in \{0, 1\}$ evolves as a persistent Markov chain with $\Pr(s_{t+1} = s_t) = 1 - \varepsilon$; the parameter $\varepsilon \in [10^{-3}, 10^{-1}]$ controls the correlation length $\ell \sim 1/\varepsilon$. Observations $o_t \in \{A, B\}$ are aliased: $\Pr(o_t = A \mid s_t = 0) = 0.5 + \delta$, $\Pr(o_t = A \mid s_t = 1) = 0.5 - \delta$, with $\delta = 0.05$ (mutual information $I(O; S) \approx 0.007$ bits). Reward: $r_t = 1$ if $a_t = s_t$, 0 otherwise.

Agents. All agents use the true model parameters and compute exact Bayesian posteriors; the only difference is how many observations each agent retains.

- **Markov-k** ($k \in \{1, 2, 4, 8\}$): runs an exact Bayes filter over the most recent k observations (sliding window, uniform prior at each window start); acts by MAP.
- **Memory (Bayes filter)**: maintains the full belief state $b_t = \Pr(s_t = 1 \mid o_{1:t})$ via the exact predict–update cycle over all past observations; acts by MAP.

Parameters.

Quantity	Value	Role
T	100,000	horizon per trial
Seeds	10	independent replications
δ	0.05	observation asymmetry
k	$\{1, 2, 4, 8\}$	Markov window sizes
ε	logspace(10^{-3}, 10^{-1}, 15)	transition noise grid
Burn-in	5,000	discarded steps

8.2 Results

Figure 2 shows the two key signatures.

Result 1: finite-window performance saturation (Figure 2a). The average reward $\bar{R}$ of the Bayes filter (memory agent) increases monotonically with correlation length $\ell = 1/\varepsilon$, while each Markov-k agent saturates at a distinct ceiling. The ceilings are ordered: $k = 1$ (lowest) through $k = 8$ (highest), and all fall below the memory agent for $\ell \gtrsim 20$. This is a property of the specified aliased HMM and agent classes, not a consequence of the thermodynamic ceiling.

Result 2: Memory advantage (Figure 2b). The gap $\Delta\bar{R} = \bar{R}_{\text{mem}} - \bar{R}_{\text{Markov-}k}$ increases monotonically with ℓ, and is larger for smaller k. Shaded bands show 95% confidence intervals across 10 seeds. The Markov-1 and Markov-2 curves nearly overlap at small ℓ, reflecting the fact that short observation windows provide negligible additional information in this aliasing regime—a consistency check, not a deficiency.

8.3 Scope of This Demonstration

These simulations illustrate finite-window saturation under the stated model class; they do not constitute a thermodynamic test.

This demonstration **does** show:

1. A reproducible regime in which finite-order Markov agents exhibit a performance ceiling while a memory-carrying (Bayes filter) agent improves.

2. The performance gap widens with correlation length and tightens with window size in this HMM. This is conceptually related to retained history but is not the quantity in Theorem 21.

This demonstration does **not** show:

1. Universality across environments, observation models, or agent architectures. The model uses a two-state HMM with binary aliased observations.

2. Tight constants or the functional form of the ceiling boundary $\ell_c(k)$.

3. That the Bayes filter is optimal among all possible memory-carrying agents.

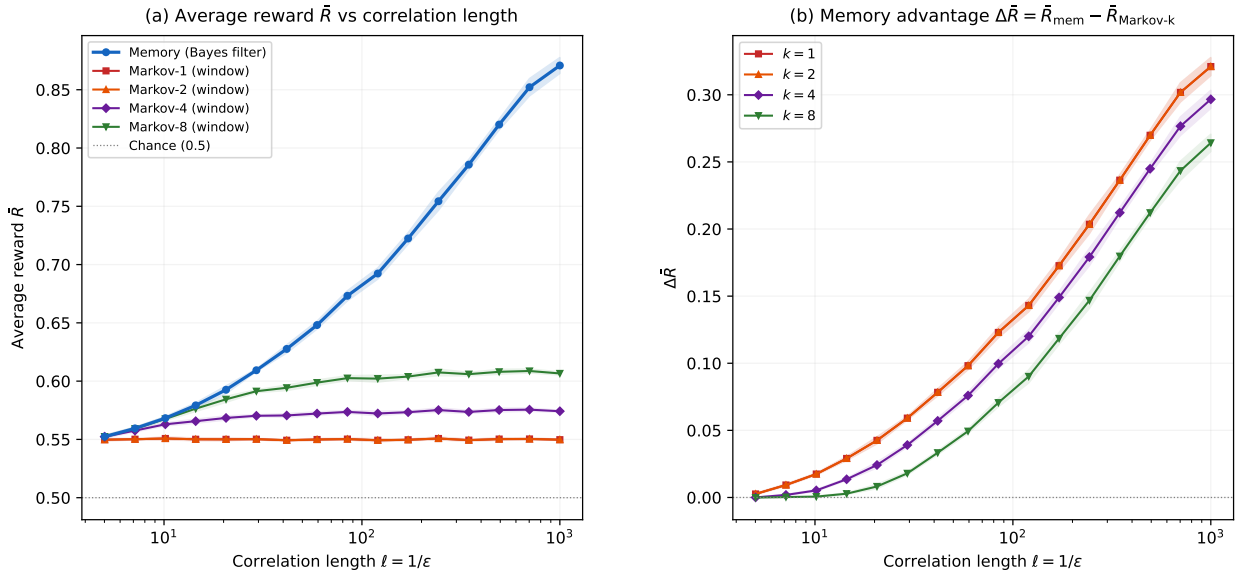


Figure 2: **Finite-window saturation and Bayes-filter advantage.** $T = 100{,}000$, 10 seeds, 95% CI bands. **(a)** Average reward $\bar{R}$ vs correlation length $\ell = 1/\varepsilon$. The Bayes filter (blue, bold) rises monotonically; Markov-k agents saturate at k-dependent ceilings. **(b)** Performance gap $\Delta\bar{R} = \bar{R}_{\text{mem}} - \bar{R}_{\text{Markov-}k}$ increases with ℓ; smaller k yields a larger gap.

Reproducibility. The complete simulation is a self-contained Python script (`paper1_markov_ceiling_demo.py`, $\sim$560 lines, requiring only NumPy and Matplotlib) with fixed random seeds. All figures in this section can be reproduced by executing the script. The following files are included in the supplementary archive:

- `paper1_markov_ceiling_demo.py` — simulation script
- `fig_paper1_markov_ceiling.pdf` — Figure 2
- `markov_ceiling_data.csv` — raw sweep data
- `markov_ceiling_boundary.csv` — extracted ceiling boundaries $\ell_c(k)$

9 Discussion

9.1 Summary of Results

Result	Statement	Sec.
Restricted autonomous ceiling	$\mathcal{S} \leq 0$ for a fixed-Hamiltonian Davies semigroup	3
Initial-resource balance	$\beta\mathcal{S} = -\Delta I - \Delta D_{\mathrm{KL}} - \beta\Delta\langle H_{\mathrm{ctrl}}\rangle - \beta\Delta\langle H_{\mathrm{int}}\rangle$	4
Quantitative demo	Spin-boson: $\beta\mathcal{S} \approx +0.093 > 0$ at first backflow revival (Fig. 1, Table 1)	6
Memory/record interface	operational influence relation plus a conditional record test	5
Archive overload	conditional on raw storage, finite capacity, and irreversible reset	7
Numerical demo	finite-window saturation in an aliased HMM (Fig. 2)	8

9.2 What This Paper Does and Does Not Show

This paper shows:

1. Under an autonomous fixed-Hamiltonian Davies semigroup, the survival functional $\mathcal{S} \leq 0$ (Theorem 13).

2. For any initial state (product or correlated), the survival functional satisfies the exact identity $\beta\,\mathcal{S} = -\Delta I(S{:}E) - \Delta D_{\mathrm{KL}}(\rho_E \| \rho_E^{\mathrm{th}}) - \beta\,\Delta\langle H_{\mathrm{ctrl}}\rangle - \beta\,\Delta\langle H_{\mathrm{int}}\rangle$ (Theorem 21). Under autonomous evolution, when pre-existing system–environment correlations are consumed ($\Delta I < 0$), $\mathcal{S} > 0$ is achievable, bounded by the initial correlation budget (Corollary 24).

3. A quantitative spin-boson example illustrates: $\beta\,\mathcal{S} \approx +0.093 > 0$ at the first non-Markovian revival (Section 6).

4. The memory-influence relation and HAFF record ordering define a conditional model test, not a proved embedding (Conjecture 35).

5. Landauer supplies a conditional reset cost; a raw finite archive can overload under the implementation assumptions in Remark 44.

6. A computational demonstration exhibits finite-window saturation in a two-state HMM with aliased observations (Section 8, Figure 2).

This paper does not show:

1. That memory or non-Markovianity is necessary or sufficient for persistence.

2. That non-Markovian systems optimize a larger survival supremum than all Markovian systems.

3. That driven or measurement-feedback Markovian agents obey the restricted autonomous ceiling (Remark 14).

4. That the specific form of the optimal memory kernel can be derived from first principles without specifying the environment.

5. That memory implies or requires consciousness.

6. That storage or maintenance has a universal Landauer cost, or that compression and symmetry breaking are thermodynamic necessities.

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